*1. Find f[g(x)] and g[f(x)].

$$f(x) = \frac{x}{10} + 3, \quad g(x) = 6x - 5$$

f[g(x)] = ______

g[f(x)] = ______

*2. Find f(x) and g(x) such that $h(x) = (f \circ g)(x)$ and $g(x) = 2 - 5x$.

$h(x) = (2 - 5x)^3 - 3(2 - 5x)^2 + 2(2 - 5x) - 1$

f(x) = ______

3. A manufacturer of a particular item has monthly fixed costs of $700 and variable costs of $20 per item, and it sells the items for $45 per item.
a. Write a function that models the monthly profit P from the production and sale of x units of the item.
b. What is the profit if 100 items are produced and sold in 1 month?
c. At what rate does the profit grow as the number of items increases?

The function that models the monthly profit P from the production and sale of x units of the item is P = ______.
(Simplify your answer. Do not factor.)

The profit of 100 items produced and sold in 1 month is $______.
(Simplify your answer.)

The rate at which the profit grows is $______ per item.
(Simplify your answer.)

4. Suppose the weekly cost for the production and sale of bicycles is $C(x) = 23x + 4488$ dollars and that the total revenue is given by $R(x) = 86x$ dollars, where x is the number of bicycles.

a. Write the equation of the function that models the weekly profit from the production and sale of x bicycles.
b. What is the profit on the production and sale of 150 bicycles?
c. Write the function that gives the average profit per bicycle.
d. What is the average profit per bicycle if 150 are produced and sold?

a. P(x) = ______

b. The profit is $______.

c. $\overline{P}(x)$ = ______

d. The average profit per bicycle is $______. (Round to the nearest cent as needed.)